Project Proposal: MATH 4090 – Mathematical Modeling

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Title

A Stochastic Exposure-Threshold Model for the Spread of Information on Social Media

1 Introduction

Social media platforms have become a dominant medium for the spread of information, opinions, and cultural trends. Content such as posts, short videos, and memes can gain visibility through a combination of direct sharing, reposting, and algorithmic amplification. Platforms like Instagram illustrate this process clearly: users may encounter the same piece of content multiple times through their feed, through reposts made by friends, or through platform features that highlight what others have interacted with or reposted.

Recent reposting and resharing features further increase the visibility of content by allowing users to redistribute posts in ways that are not always explicit "shares" but still push content into the feeds of connected users (REF1). As a result, exposure is often cumulative rather than singular.

In practice, users do not typically share content immediately upon first exposure. Instead, repeated exposure can increase familiarity and perceived relevance, making users more likely to engage or share at a later time. This suggests that information diffusion on social media depends on both **repeated exposure mechanisms** and **individual variability in decision-making**.

2 Literature Review

The mathematical modeling of rumor and information spread has been studied extensively. One of the earliest and most influential models is the *Daley–Kendall stochastic rumor model*, which divides individuals into ignorants, spreaders, and stiflers (REF2). A closely related foundational model is the *Maki–Thompson rumor model*, which incorporates the eventual decline of spreading activity (REF3).

More recent research has extended these frameworks to reflect modern environments. Several models emphasize **repeated exposure**, where individuals only become spreaders after encountering information multiple times (REF4). Other studies focus on **limited**

attention and competition among pieces of information, showing that popular content may experience rapid decline due to attention fatigue (REF5).

This project builds on the existing literature by proposing a stochastic model that combines repeated exposure thresholds with probabilistic sharing and attention decay.

3 Model

3.1 Model Structure and Conceptual Framework

The aim of the model is to represent how information spreads when users require multiple exposures before deciding to share. Users begin with no exposure and accumulate it over time. Once a user reaches the threshold m, they become eligible to share. However, sharing is probabilistic, reflecting individual variability. After sharing, users gradually lose interest (decay).

3.2 Mathematical Description of the Model

We consider a fixed population of N users and discrete time steps $t = 0, 1, 2, \ldots$.

Variables:

- $I_k(t)$: Number of users exposed exactly k times but not sharing ($k = 0, 1, \ldots, m-1$).
- $S(t)$: Number of users actively sharing or reposting.
- $R(t)$: Number of users who are saturated and will not share further.

Exposure Probability: At each time step, a non-sharing user receives an exposure with probability p_t:

$$p_t = 1 - \exp\left(-\beta \frac{S(t)}{N}\right) \tag{1}$$

where $\beta > 0$ is the exposure intensity parameter (REF1).

Transition Rules: Each exposure increases a user's count by one. Users in class I_{m-1} who receive an exposure will:

- Share with probability ρ.
- Become saturated with probability $1 - \rho$.

Active sharers stop contributing with probability γ per time step (REF5).

Expected Update Equations:

$$I_0(t+1) = I_0(t) - p_t I_0(t) \tag{2}$$
$$I_k(t+1) = I_k(t) + p_t I_{k-1}(t) - p_t I_k(t), \quad k = 1, \ldots, m-2 \tag{3}$$
$$\Delta_{share}(t) = \rho p_t I_{m-1}(t) \tag{4}$$
$$\Delta_{stifle}(t) = (1-\rho) p_t I_{m-1}(t) \tag{5}$$
$$\Delta_{decay}(t) = \gamma S(t) \tag{6}$$
$$S(t+1) = S(t) + \Delta_{share}(t) - \Delta_{decay}(t) \tag{7}$$
$$R(t+1) = R(t) + \Delta_{stifle}(t) + \Delta_{decay}(t) \tag{8}$$

Population Constraint:

$$N = \sum_{k=0}^{m-1} I_k(t) + S(t) + R(t) \tag{9}$$

3.3 Table 1: Model Variables and Parameters

Symbol	Description
N	Total population size
$I_k(t)$	Users exposed k times but not sharing
$S(t)$	Active sharers / reposters
$R(t)$	Saturated (non-sharing) users
m	Exposure threshold
β	Exposure intensity parameter
ρ	Probability of sharing once eligible
γ	Attention decay rate

4 Proposed Analysis

The model will be analyzed using probabilistic reasoning and Monte Carlo simulations. Key quantities of interest include:

- The final reach of the content.
- The maximum number of active sharers and time to peak activity.
- The probability of a large-scale outbreak.

Sensitivity analysis will be conducted by varying β, ρ, γ, and m to examine how they influence diffusion outcomes.

5 References

REF1 Meta Platforms Inc., Instagram product updates on content visibility.

REF2 Daley, D. J. and Kendall, D. G., *Stochastic Rumours*, IMA Journal of Applied Mathematics.

REF3 Maki, D. P. and Thompson, M., *Mathematical Models and Applications*.

REF4 Studies on repeated-exposure rumor models and adoption thresholds.

REF5 Weng et al., models of limited attention and meme competition.

REF6 Network-based and simulation studies on information diffusion.